

Title: Overcoming Catastrophic Forgetting: Synaptic Plasticity vs. Parameter Overwriting

Target Audience: Intermediate Data Scientists

The Design Pressure: The Stability-Plasticity Trade-Off

As artificial intelligence transitions from static models to active systems, the ability to engage in continual learning has become a critical design pressure. Deployed models must adapt to new tasks, environments, and constraints without erasing earlier capabilities. However, standard neural networks, particularly Transformer-based Large Language Models (LLMs), suffer from a mathematical limitation known as catastrophic forgetting. When a deployed Transformer is fine-tuned to learn a new rule, its optimisation process overwrites the fixed parameters that encoded its prior knowledge. It fails the stability-plasticity trade-off: it shows plasticity (learning the new task) at the absolute expense of stability (forgetting the old task).

The Technical Shift: Architecture-Native Session Memory

To solve this, memory must be decoupled from permanent parameter weights. Continual learning requires architecture-native session memory, where adaptation happens in a recurrent state rather than in foundational weights. This shift allows a model to temporarily strengthen connections, turning the network's wiring into working memory, without causing destructive interference to its core capabilities.

BDH and BDH-CQ Integration

Pathway's Dragon Hatchling (BDH) provides a primary case study in addressing this flaw. BDH is a brain-inspired post-transformer architecture where reasoning and memory share one computational fabric. Instead of relying on a growing key-value cache or standard parameter updates, BDH reformulates attention as synaptic memory.

Through Hebbian writes, recent activity temporarily strengthens connections,

$$W_{new} = (1 - \gamma)W_{old} + \eta(x \cdot y)$$

allowing the model to adapt dynamically. Furthermore, BDH-CQ (a later system in the family) shows this by learning from sparse demonstrations and reasoning without a written chain of thought, successfully adapting to unseen evaluation tasks entirely through recurrent state rather than via parameter updates.

Comparative Analysis

Standard Transformers: Rely on durable parameter updates for learning new tasks post-deployment. High catastrophic forgetting. Context is limited by the KV cache size, preventing long-horizon evolving states.

State-Space Models (SSMs): Compress sequences into an evolving state. Highly efficient but still require permanent parameter adjustments for distinct task fine-tuning, leaving them vulnerable to forgetting over discrete, separated training sessions.

BDH / Synaptic Plasticity Models: Distinguish durable parameter learning from temporary adaptation through context. High plasticity for immediate tasks without sacrificing the stability of the foundation.

Limitations and Unanswered Questions

While synaptic memory resolves immediate test-time adaptation without forgetting, a major limitation stays in long-term knowledge consolidation. It is currently unclear how a model effectively transfers incredibly useful "fast" temporary state into "slow" durable parameters over extended periods without eventually triggering interference. Furthermore, independent verification is constrained, as specific architectural scales and layer dimensions of BDH remain proprietary.

Primary References

1. Wang, L., et al. (2024). "A Comprehensive Survey of Continual Learning: Theory, Method and Application." (Provides foundational evidence on the stability-plasticity trade-off and catastrophic forgetting).
2. Luo, Y., et al. (2023). "Catastrophic Forgetting in Large Language Models." (Demonstrates that standard Transformer architectures inherently overwrite capabilities during continual fine-tuning).
3. Pathway. (2026). "Dragon Hatchling (BDH-CQ) Technical Report." (Validates test-time adaptation and skill acquisition from demonstrations without parameter updates).